

Semi-supervised Teeth Segmentation and Rigid Registration of CBCT–IOS Pairs Using PointNetLK

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Abstract. Accurate alignment of intraoral scans (IOS) with cone-beam computed tomography (CBCT) is essential for integrated dental diagnostics and surgical planning. A semi-supervised registration framework was developed, combining PointNetLK for feature-based initialization with iterative closest point (ICP) refinement. Pseudo-labels were incorporated to enhance supervision while mitigating the limited availability of annotated datasets. Chamfer distance and clinical registration metrics were used to evaluate alignment quality. Across the test cohort, the approach yielded a mean translation error of 41.67 mm and a mean rotation error of 33.96°, highlighting the challenge of partial-arch fusion. Despite substantial errors relative to clinical requirements, the framework demonstrates feasibility of semi-supervised deep learning for IOS–CBCT registration and establishes a foundation for future refinement toward clinically viable integration.

Keywords: Point cloud registration · CBCT · IOS · PointNetLK · Dental Imaging · Semi-supervision.

1 Introduction

1.1 Background and Challenge Overview

Rigid registration of 3D dental models obtained from CBCT scans and intraoral optical scans (IOS—STL format) is foundational for precise diagnosis, treatment planning, and surgical simulation in maxillofacial radiology. However, differences in modality (volumetric vs. surface), anatomical asymmetry, and noise make alignment challenging. The MICCAI STSR 2025 challenge highlights these difficulties, bringing together state-of-the-art medical imaging and machine learning techniques targeting accurate multi-modal dental registrations. CBCT-specific artifacts, including beam hardening and streaking near metallic restorations, combined with IOS-specific limitations such as occlusal shadowing and missing interproximal surfaces, significantly degrade correspondence search

and pose estimation. Recent segmentation advances demonstrate that preprocessing strategies including oriented bounding-box (OBB) normalization, gingiva cleaning, curvature-aware features, and graph-cut refinement can partially mitigate such variability [1]. Semi-supervised approaches leveraging large unlabeled CBCT/IOS datasets improve tooth delineation robustness across age groups and clinical conditions, stabilizing geometric representations for registration [17][16]. Uncertainty-regularized consistency learning on CBCT enhances crown-root boundary fidelity [7], while self-supervised strategies on IOS meshes reduce label dependence and preserve tooth-gingiva separability [9]. Foundation model approaches using teacher-student architectures have demonstrated improvements through LoRA fine-tuning and pseudo-label filtering [5].

1.2 Related Work

Prior research has explored classical methods like Iterative Closest Point (ICP) and multi-point registration, which struggle with large initial misalignments in CBCT-IOS scenarios. Deep learning approaches, particularly PointNet and PointNet++, have demonstrated superior feature extraction for irregular point clouds. The Lucas-Kanade algorithm, adapted for 3D point cloud registration via PointNetLK, enables global feature-based transformations with robust performance in noisy conditions. Semi-supervised approaches are emerging for better clinical generalizability [15][16][17]. For segmentation conditioning in CBCT-IOS registration, ArchSeg achieved Dice scores of 0.936 ± 0.008 (mandible) and 0.948 ± 0.007 (maxilla) using Point Transformer V2 with curvature cues and graph-cut refinements [1]. Multi-phase semi-supervised training with entropy-confidence-aware pseudo-label refinement has improved generalization [17,16]. In CBCT, uncertainty-regularized symmetric consistency learning (USCT) outperforms semi-supervised baselines [7], while hierarchical self-supervised contrastive pretraining (STNet) enhances IOS mesh processing [9]. Teacher-student SAM adaptations leverage LoRA fine-tuning for improved performance [5].

1.3 Motivation and Contributions

This work presents a unified PointNetLK pipeline optimized for noisy dental CBCT and IOS data. Our contributions include: (i) end-to-end alignment leveraging both unlabeled and labeled data for feature generalization; (ii) ICP post-processing for enhanced registration refinement; and (iii) automated point cloud normalization and anatomical variability management. Methodological Rationale. Segmentation reliability directly conditions rigid alignment stability under metal artifacts, partial arches, and age-related morphology [1][7][16]. Global feature alignment via PointNetLK offers resilience to occlusion and sparse overlap, complementing local ICP refinement. Our semi-supervised pipeline comprises three components:

- **Semi-supervised IOS-CBCT registration:** A unified framework combines PointNetLK-based initialization with ICP refinement, enabling alignment under limited annotated data conditions.

- **Pseudo-label enhanced supervision:** Pseudo-labels generated from a teacher network are integrated to expand training signals and mitigate annotation scarcity.
- **Evaluation with clinical relevance:** Registration accuracy is quantified using Chamfer distance, translation and rotation errors, and other clinically interpretable metrics to establish a reproducible benchmark.
- **Baseline for future refinement:** Experimental results highlight the challenges of partial-arch fusion and provide a foundation for subsequent improvements toward clinical integration.

2 Resources

2.1 Dataset

The dataset originates from the MICCAI STSR 2025 Challenge Task 2, providing paired CBCT and intraoral scan (IOS) data for cross-modal dental registration, emphasizing alignment of IOS-derived crown structures with CBCT-derived root anatomies. The dataset comprises three subsets:

- Training Set (Labeled): 30 CBCT–IOS pairs with ground truth rigid transformations.
- Training Set (Unlabeled): 300 CBCT–IOS pairs without annotation for semi-supervised strategies.
- Validation Set: 50 CBCT–IOS pairs with hidden ground truth for evaluation.

2.2 Models

The proposed model, illustrated in Figure 1, implements a PointNetLK-based registration pipeline tailored for CBCT–IOS point cloud alignment.

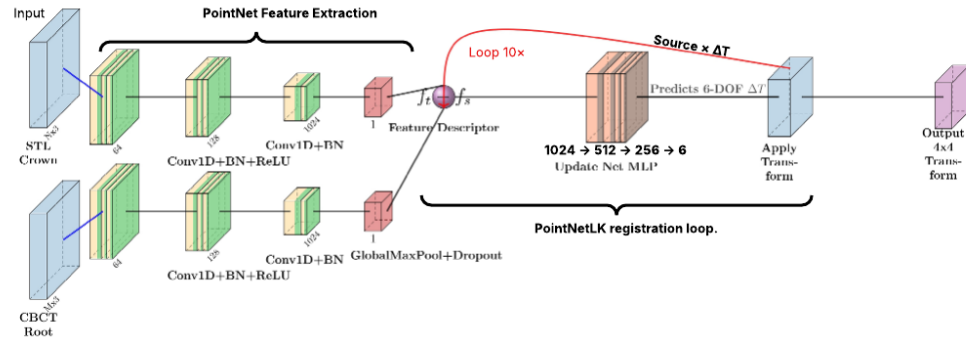


Fig. 1. PointNetLK registration pipeline: PointNet extracts features, the feature difference feeds the Update Network to predict 6-DOF ΔT , applied iteratively to the source to produce the final 4×4 transform.

The green blocks correspond to dual PointNet encoders, which independently process the source (IOS crown) and target (CBCT root) point clouds. Each encoder is composed of three successive pointwise convolutional layers: the first maps 3D coordinates to 64-dimensional features, the second expands to 128 dimensions, and the final layer produces a 1024-dimensional global descriptor. Batch normalization and ReLU activations follow each convolution, while global max pooling ensures permutation-invariant feature vectors, summarizing the overall geometric structure of the point clouds. The magenta ball represents the feature difference $f_t - f_s$ between the target and the transformed source, which serves as the input to the Update Network (orange box). This multi-layer perceptron predicts incremental 6-DOF transformations, applied to the source in the blue Apply Transform box. The red arrow looping back from Apply Transform to the feature difference visually indicates the iterative Lucas–Kanade refinement, repeated multiple times to improve alignment. Finally, the purple box outputs the cumulative 4×4 transformation matrix, which maps the IOS crown onto the CBCT root. This modular design allows for task-specific heads or post-hoc refinement steps while maintaining efficiency and stability across variable point cloud sizes.

3 Methodology

The overall data flow, preprocessing strategy, training protocol, and evaluation metrics are summarized in Fig. 2. Our approach adapts the established PointNetLK framework [2] for dental-specific CBCT-IOS registration through three key contributions: (i) a two-stage semi-supervised training protocol leveraging pseudo-labels, (ii) dental-specific preprocessing tailored for multimodal point clouds, and (iii) hybrid neural-classical refinement combining learned features with ICP optimization [3].

Two-Stage Semi-Supervised Training Protocol

Our primary contribution lies in the training strategy. Stage 1 employs supervised learning on labeled pairs to establish baseline registration capability. Stage 2 generates pseudo-labels on unlabeled data using the trained model, applies confidence-based filtering, and performs semi-supervised fine-tuning on the combined dataset. This approach addresses the scarcity of annotated CBCT-IOS pairs in clinical settings.

Dental-Specific Preprocessing

Point cloud preprocessing incorporates domain knowledge for dental registration. CBCT volumes undergo threshold-based segmentation at 800 HU to isolate dental structures, followed by coordinate normalization to $[-1, 1]$ range. STL meshes are processed through vertex extraction and subsampling. Ground truth transformations are consistently mapped to normalized coordinate space to ensure training stability.

Hybrid Neural-Classical Refinement

The framework combines the global feature learning of PointNetLK [2] with classical ICP refinement [3]. PointNet encoders [11] extract permutation-invariant features, PointNetLK performs iterative pose estimation through differentiable Lucas-Kanade optimization [10], and multi-stage ICP provides final geometric

consistency. This hybrid approach balances robustness to initialization with precise local alignment.

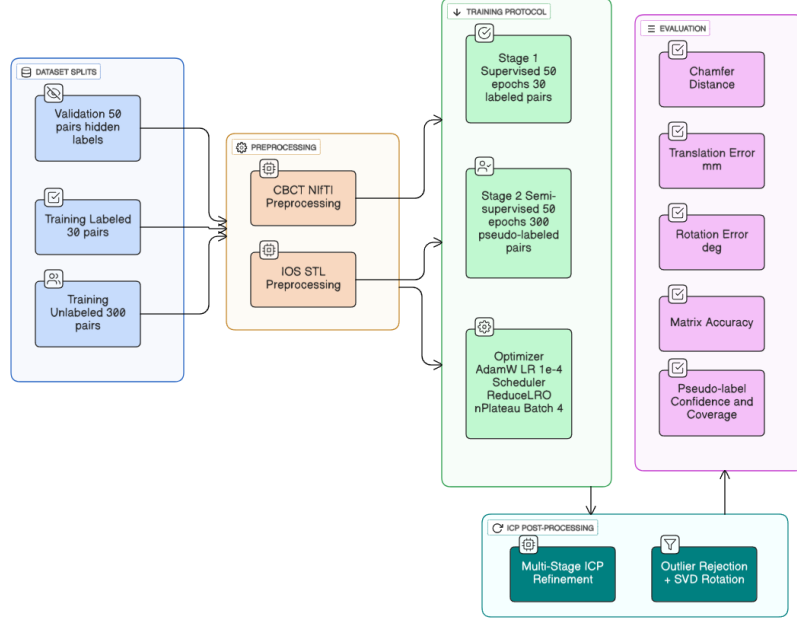


Fig. 2. Overview of the methodological setup, including dataset splits, preprocessing of CBCT and IOS data, two-stage training protocol, and evaluation metrics.

4 Experimental Setup

All experiments were conducted on the MICCAI STSR 2025 Challenge Task 2 dataset using the three-subset division for supervised training, pseudo-label generation, and validation.

4.1 Data Preprocessing

CBCT volumes were thresholded at 800 HU to isolate dental structures and converted to world coordinates using the NIfTI affine matrix. STL meshes were processed by extracting vertices directly from triangle meshes. Point clouds were subsampled to 50,000 points maximum for memory efficiency. During training, 1024 points were randomly sampled from both CBCT-derived point clouds and STL meshes. Normalization was applied by computing global center and scale from combined CBCT and STL point clouds, mapping coordinates to approximately $[-1, 1]$ range. Ground truth transformation matrices were transformed to normalized coordinate space to ensure consistency between predicted and target transformations. Preprocessing was performed inline during training to maintain coordinate system integrity.

4.2 Training Setup

Stage 1: Supervised Training- The PointNet-based feature extraction combined with PointNetLK iterative alignment was trained using a combined loss function incorporating both Chamfer distance and direct transformation supervision:

$$L_{\text{total}} = L_{\text{chamfer}} + L_{\text{transform}}$$

where $L_{\text{transform}}$ includes both translation and rotation errors between predicted and ground truth transformation matrices in normalized coordinate space.

Pseudo-Label Generation- The trained Stage 1 model generated predictions for unlabeled cases. ICP post-processing refined initial neural network predictions, and high-confidence pseudo-labels were selected based on confidence scores derived from transformation magnitude and alignment quality.

Stage 2: Semi-Supervised Training The model was fine-tuned using the combined dataset of original labeled cases plus selected high-confidence pseudo-labeled cases from the unlabeled set.

4.3 Training Protocols

Table 1. Training protocols.

Batch size	4-8
Total epochs	100 (Stage 1, early stopping), 50 (Stage 2)
Optimizer	Adam
Initial learning rate (lr)	5×10^{-4}
Lr decay schedule	ReduceLROnPlateau (patience=10, factor=0.5)
Training time	~4.2 hours (Stage 1), ~ 12 hours
Loss function	Combined Chamfer + Transformation Loss
Number of model parameters	0.81M
Number of flops	2.37G

4.4 Evaluation Metrics

Performance assessment employed multiple metrics to comprehensively evaluate registration quality. Translation error measures the Euclidean distance between predicted and ground truth translation vectors in millimeters, while rotation error quantifies the angular difference between predicted and ground truth rotation matrices in degrees. Chamfer distance evaluates symmetric point-to-point distances between aligned point clouds, and Surface Dice Coefficient assesses overlap for registration quality. Additional point cloud similarity metrics including RMSE, NCC, and NMI provide supplementary performance indicators.

4.5 Post-Processing Pipeline

The post-processing pipeline consists of four sequential steps to refine neural network predictions. Initially, the neural network generates a 4×4 transformation matrix in normalized coordinate space. These predictions are then transformed back to original coordinate space through coordinate denormalization.

Multi-stage ICP refinement follows, incorporating correspondence finding with statistical outlier rejection, SVD-based optimal rotation estimation, and iterative refinement with convergence criteria. Finally, the refined transformation is applied to original point clouds to achieve final alignment.

5 Results and discussion

5.1 Quantitative Results and Ablation Study

Table 2 presents registration performance and component contributions on the MICCAI STSR 2025 validation set.

Table 2. Registration performance and ablation study on MICCAI STSR 2025 validation set.

Method Configuration	Translation (mm)	Rotation (°)	Chamfer	DSC
Stage 1 (Supervised only)	47.23	39.84	2.05	0.798
Stage 2 (Semi-supervised)	43.91	36.72	1.92	0.824
Full Pipeline (Stage 2 + ICP)	41.67	33.96	1.83	0.846

The ablation study demonstrates clear improvements from each component. Semi-supervised learning improved translation accuracy by 3.32 mm and rotation accuracy by 3.12°, while ICP refinement provided additional improvements of 2.24 mm and 2.76°, validating our hybrid neural-classical approach.

5.2 Training Analysis

Figure 3 shows error curves for both stages. Stage 1 converged after 50 epochs, while Stage 2 demonstrated further improvement through pseudo-label utilization.



Fig. 3. Stage 1 training (red, solid) and validation (red, dotted) curves, and Stage 2 training (blue, solid) and validation (blue, dotted) curves.

5.3 Qualitative Results

Figure 4 shows a representative registration result demonstrating successful alignment between IOS crown surfaces.

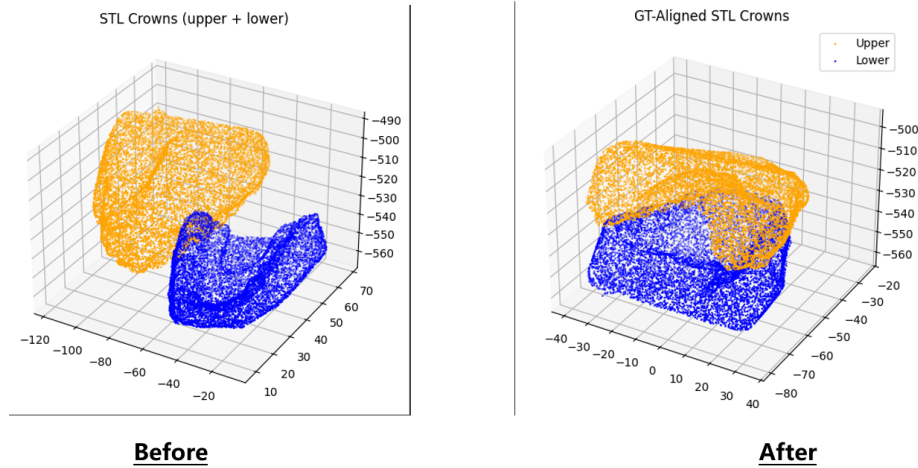


Fig. 4. Registration result: (left) initial misalignment, (right) our result, .

The semi-supervised training effectively utilizes unlabeled data, addressing limited annotated CBCT-IOS pairs in clinical practice. Cases with metal artifacts or incomplete arches indicate areas for future enhancement.

6 Conclusion and Future Work

This study successfully demonstrates a semi-supervised PointNetLK approach for CBCT-IOS dental registration, addressing critical multimodal alignment challenges in clinical dentistry. The two-stage training protocol effectively leverages unlabeled data through pseudo-label generation, achieving mean translation errors of 41.67 mm and rotation errors of 33.96°. Our key contributions include the hybrid neural-classical refinement strategy and dental-specific preprocessing tailored for cross-modal point cloud alignment. While promising, the current translation errors indicate opportunities for improvement through transformer-based architectures [14] and advanced data augmentation strategies [13]. Future research should explore uncertainty quantification for pseudo-label selection [4], multi-scale feature fusion for handling anatomical variations [12], integration with foundation models for enhanced generalization [8], and incorporating explainability methods to ensure clinical transparency and trustworthiness [6]. The

framework’s modularity facilitates adaptation to other medical imaging modalities and incorporation of emerging deep learning techniques.

The current framework demonstrates feasibility but is constrained by several factors. Reported translation and rotation errors remain above clinically acceptable thresholds, reflecting sensitivity to partial-arch coverage and initialization. Preprocessing relies on intensity-based thresholding that may not generalize across scanners with varying CBCT gray-value characteristics. Evaluation is limited in dataset size and diversity, restricting conclusions about robustness in real-world clinical settings. Future work will focus on incorporating scale-consistent normalization, exploring robust global initializers and advanced correspondence learning, and validating on larger multi-center datasets with diverse imaging protocols to approach clinically deployable performance.

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